

MATH3881/5881 — Statistical Machine Learning Theory

Sample Short-Answer Questions: Solutions

(Mid-Term Exam, Term 2 2026)

Worked solutions to the sample short-answer questions. Numbering and part labels match the question sheet.

Sample S1. Pooling and ReLU: swap or not?

(a) Let $x^* = \max_{i,j} x_{ij}$. Since ReLU is non-decreasing, $\text{ReLU}(x_{ij}) \leq \text{ReLU}(x^*)$ for every (i, j) , with equality at the maximising entry. Hence

$$\max_{i,j} \text{ReLU}(x_{ij}) = \text{ReLU}(x^*),$$

which is the claimed identity.

(b) It fails. Take $x = \begin{pmatrix} -4 & 4 \\ 0 & 0 \end{pmatrix}$: then

$$\text{ReLU}(\text{avgpool}(x)) = \text{ReLU}(0) = 0, \quad \text{avgpool}(\text{ReLU}(x)) = \frac{1}{4}(0 + 4 + 0 + 0) = 1.$$

The two sides differ, so the identity does not hold for average pooling.

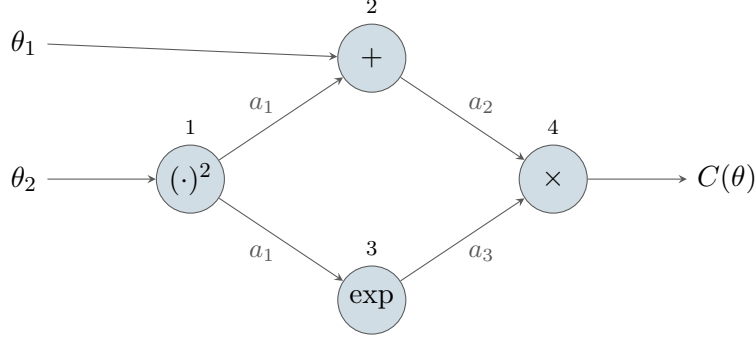
(c) By (a) both orders give the same output, so only cost differs. Pooling first shrinks the feature map by a factor of 4, so ReLU is applied to $4\times$ fewer entries; the pooling cost is identical in both orders. Pool-then-ReLU is cheaper by a factor of 4 in ReLU evaluations.

Sample S2. Reverse-mode differentiation on a computational graph.

(a) Decomposition:

$$a_1 = \theta_2^2, \quad a_2 = \theta_1 + a_1, \quad a_3 = e^{a_1}, \quad C = a_2 \cdot a_3.$$

The computational graph (nodes are elementary operations, labelled 1 to 4; outgoing edges carry the primals):



The node with two outgoing edges is node 1 (the squaring node): its primal a_1 feeds both node 2 and node 3.

(b) At $(1, 1)$: $a_1 = 1$, $a_2 = 2$, $a_3 = e$, $C = 2e$.

(c) C depends on a only through $u_1, \dots, u_m$, so the multivariate chain rule gives

$$\zeta_a = \frac{\partial C}{\partial a} = \sum_{j=1}^m \frac{\partial C}{\partial u_j} \frac{\partial u_j}{\partial a} = \sum_{j=1}^m \zeta_{u_j} \frac{\partial u_j}{\partial a}.$$

(d) Starting from $\zeta_C = 1$:

$$\zeta_{a_2} = a_3 = e, \quad \zeta_{a_3} = a_2 = 2,$$

and, summing over a_1 's two consumers as in (c),

$$\zeta_{a_1} = \zeta_{a_2} \cdot 1 + \zeta_{a_3} \cdot e^{a_1} = e + 2e = 3e.$$

For the inputs,

$$\zeta_{\theta_1} = \zeta_{a_2} \cdot 1 = e, \quad \zeta_{\theta_2} = \zeta_{a_1} \cdot 2\theta_2 = 6e.$$

Hence $\nabla C(1, 1) = (e, 6e)$. Direct check: $\frac{\partial C}{\partial \theta_2} = 2\theta_2 e^{\theta_2^2} + (\theta_1 + \theta_2^2) e^{\theta_2^2} \cdot 2\theta_2 = 2e + 4e = 6e$ at $(1, 1)$.

Sample S3. Exact representation by shallow ReLU networks.

(a) Take $f_\theta(x) = \text{ReLU}(x) + \text{ReLU}(-x)$, i.e., $w = (1, -1)$, $b = (0, 0)$, $\alpha = (1, 1)$, $c = 0$. For $x \geq 0$ it equals $x + 0 = |x|$; for $x < 0$ it equals $0 + (-x) = |x|$.

(b) Let $f_\theta(x) = \alpha \text{ReLU}(wx + b) + c$. If $\alpha = 0$ or $w = 0$ then f_θ is constant on $\mathbb{R}$; otherwise $f_\theta \equiv c$ on the half-line $\{x : wx + b < 0\}$. In every case f_θ is constant on some half-line, while $|x|$ is constant on none. So $N = 1$ is impossible.

(c) Each term $\alpha_i \text{ReLU}(w_i x + b_i)$ is affine except possibly at the single breakpoint $x_i^* = -b_i/w_i$ (no breakpoint if $w_i = 0$). Hence f_θ can change slope only at the at most N points $x_1^*, \dots, x_N^*$, which partition $\mathbb{R}$ into at most $N + 1$ intervals; on each interval f_θ is a sum of affine functions, hence affine.

(d) If $f_\theta = x^2$ on $[0, 1]$, then by (c) the at most $N + 1$ linear pieces of f_θ cover $[0, 1]$, so f_θ is affine on some subinterval $[u, v] \subseteq [0, 1]$ with $u < v$. But x^2 is strictly convex: $\left(\frac{u+v}{2}\right)^2 < \frac{u^2+v^2}{2}$, whereas affine functions give equality at the midpoint. Contradiction, so no finite N suffices. This does not contradict the UAT: the theorem guarantees approximation within any $\varepsilon > 0$, not exact equality.

Sample S4. Adam on an intermittent gradient.

(a) Taking expectations of the recursion, $\mathbb{E}[u^{(t+1)}] = \beta \mathbb{E}[u^{(t)}] + (1 - \beta) m$. By induction from $\mathbb{E}[u^{(0)}] = 0$:

$$\mathbb{E}[u^{(t+1)}] = \beta (1 - \beta^t) m + (1 - \beta) m = (1 - \beta^{t+1}) m.$$

Since $\beta \in [0, 1)$, $\beta^{t+1} \rightarrow 0$ and $\mathbb{E}[u^{(t+1)}] \rightarrow m$.

(b) $v^{(t)}$ and $s^{(t)}$ are exponential moving averages of $g^{(t)}$ and $(g^{(t)})^2$, whose means are $\mathbb{E}[g^{(t)}] = q g$ and $\mathbb{E}[(g^{(t)})^2] = q g^2$. By (a),

$$\lim_{t \rightarrow \infty} \mathbb{E}[v^{(t)}] = q g, \quad \lim_{t \rightarrow \infty} \mathbb{E}[s^{(t)}] = q g^2.$$

(c) With bias corrections ≈ 1 and $\varepsilon = 0$, replace $\hat{v}$ and $\hat{s}$ by their limits:

$$|\theta^{(t+1)} - \theta^{(t)}| \approx \alpha \frac{q g}{\sqrt{q g^2}} = \alpha \frac{q g}{\sqrt{q} g} = \alpha \sqrt{q}.$$

(d) The gradient-descent step magnitude is αg with probability q and 0 otherwise, so its average is $\alpha g q$. The ratio is

$$\frac{\alpha \sqrt{q}}{\alpha g q} = \frac{1}{g \sqrt{q}} \rightarrow \infty \quad \text{as } q \rightarrow 0.$$